

Step-by-Step Explanation of My Subnetting Plan

I started with the network **192.168.10.0/24**. The **/24** means the subnet mask is **255.255.255.0**. A **/24** network contains **256 IP addresses**, starting from **192.168.10.0** and ending at **192.168.10.255**.

Step 1: Decide How Many Subnets Are Needed

The question asked me to divide the network into **4 equal subnets**.

To find how many bits I need to borrow, I used this formula:

$$2^n = \text{Number of Subnets}$$

Where **n** is the number of bits borrowed.

- $2^1 = 2$ subnets (Not enough)
- $2^2 = 4$ subnets (Correct)

So, I borrowed **2 bits** from the host portion.

Step 2: Find the New Subnet Mask

The original prefix was **/24**.

After borrowing **2 bits**, the new prefix becomes:

$$/24 + 2 = /26$$

The subnet mask for **/26** is:

$$255.255.255.192$$

Step 3: Find the Number of IP Addresses per Subnet

The block size is calculated using this formula:

$$256 - 192 = 64$$

So, each subnet contains **64 IP addresses**.

Out of these:

- **1 IP is the Network Address**
- **1 IP is the Broadcast Address**
- **62 IPs are available for devices**

Step 4: Find the Four Subnets

Subnet	Network Address	Broadcast Address	Valid Host Range
1	192.168.10.0/26	192.168.10.63	192.168.10.1 – 192.168.10.62
2	192.168.10.64/26	192.168.10.127	192.168.10.65 – 192.168.10.126
3	192.168.10.128/26	192.168.10.191	192.168.10.129 – 192.168.10.190
4	192.168.10.192/26	192.168.10.255	192.168.10.193 – 192.168.10.254

Step 5: Assign IP Addresses to Devices

I selected **Subnet 1 (192.168.10.0/26)**.

The usable IP range is:

192.168.10.1 to 192.168.10.62

I did **not** use:

- **192.168.10.0** because it is the **Network Address**.
- **192.168.10.63** because it is the **Broadcast Address**.

Device	IP Address
Router (Gateway)	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5